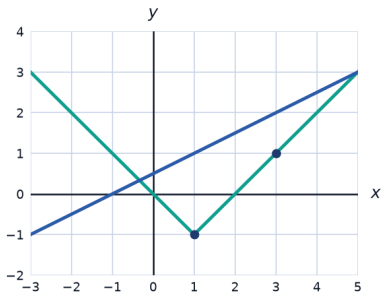
Functions & Graphs

Section 1: Function Notation & Operations

1 If $f(x) = 4x - 5$ and $g(x) = 2x^2 + 1$, what is $(f + g)(3)$?



a. 24

b. 26

c. 28

d. 30

2 If $f(x) = 3x + 2$, what is the expression for $f(2a - 1)$?

a. $6a - 1$

b. $6a + 1$

c. $6a - 3$

d. $5a + 1$

3 What is the domain of the function $h(x) = \frac{\sqrt{x-4}}{x-7}$?

a. $[4, 7) \cup (7, \infty)$

b. $[4, \infty)$

c. $(4, 7) \cup (7, \infty)$

d. $x \neq 7$

Section 2: Composite Functions & Inverses

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4 If $f(x) = 2x - 3$ and $g(x) = x^2 + 4$, what is $(f \circ g)(2)$?

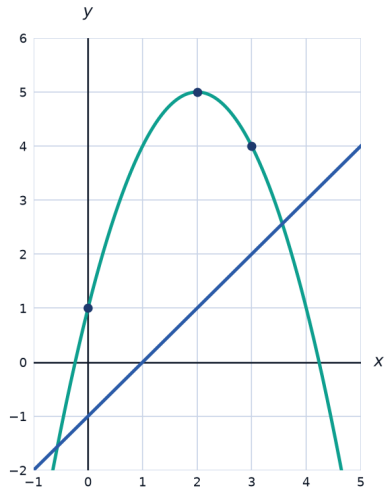
a. 13

b. 15

c. 17

d. 19

5 What is the inverse function of $f(x) = \frac{5x+3}{2}$?



a. $f^{-1}(x) = \frac{2x-3}{5}$

b. $f^{-1}(x) = \frac{2x+3}{5}$

c. $f^{-1}(x) = \frac{5x-2}{3}$

d. $f^{-1}(x) = \frac{2}{5x+3}$

6 If $f(f^{-1}(x)) = x$ for all x, and $f(4) = 9$, what is $f^{-1}(9)$?

a. 4

b. $\frac{1}{4}$

c. $\frac{1}{9}$

d. 9

Section 3: Graph Transformations

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7 How is the graph of $y = f(x - 4) + 2$ obtained from the graph of $y = f(x)$?

a. Shifted 4 units right and 2 units up

b. Shifted 4 units left and 2 units up

c. Shifted 4 units right and 2 units down

d. Shifted 4 units left and 2 units down

8 The graph of $y = x^2$ is vertically stretched by a factor of 3 and reflected across the x-axis. What is its equation?

a. $y = -3x^2$

b. $y = 3(-x)^2$

c. $y = -\frac{1}{3}x^2$

d. $y = -(3x)^2$

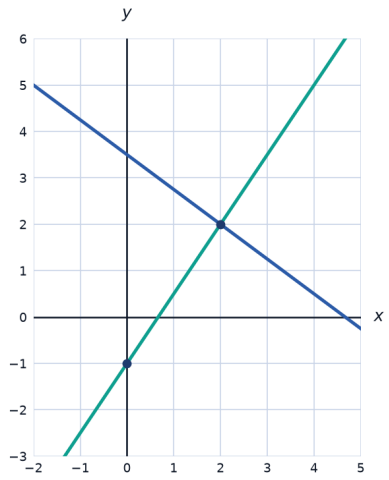
9 Which of the following functions has origin symmetry (odd function)?

a. $f(x) = x^3 - 4x$

b. $f(x) = x^2 + 5$

c. $f(x) = 3x - 2$

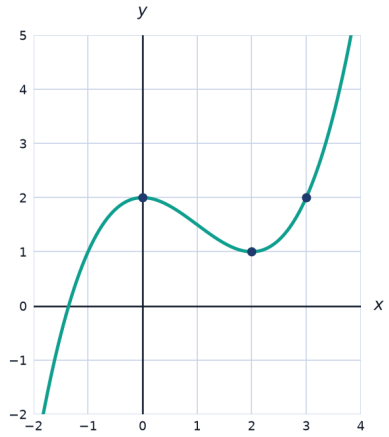
d. $f(x) = x^4 - x^2$



Section 4: Rational Graphs & Asymptotes

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10 What is the vertical asymptote of $f(x) = \frac{3x+1}{x-5}$?



a. $x = 5$

b. $x = -5$

c. $y = 3$

d. $y = 5$

11 What is the horizontal asymptote of $g(x) = \frac{4x^2-1}{2x^2+3}$?

a. $y = 2$

b. $y = 0$

c. $y = 4$

d. No horizontal asymptote

12 What are the coordinates of the hole in the graph of $h(x) = \frac{x^2-9}{x-3}$?

a. $(3, 6)$

b. $(3, 0)$

c. $(-3, 0)$

d. $(3, 3)$

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